

RadioBeat

Watching your heartbeat through the air, using only WiFi

Final Proposal · Qutuhah InnovateX 2.0

Project Title	RadioBeat: Watching Your Heartbeat Through the Air, Using Only WiFi
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Pillar	Artificial Intelligence & Machine Learning
Project site	adhvi29.github.io/radiobeat
Live monitor	adhvi29.github.io/radiobeat-monitor
Source code	github.com/adhvi29/radiobeat-monitor

1. The Problem I Wanted to Solve

At the start of this competition I spent a week looking for a problem worth solving. I landed on one silent, dangerous problem: something can go wrong with your heart while you are asleep, and most people have no way of knowing. That is the problem I set out to solve.

The condition: Atrial Fibrillation (AFib)

Normally your heart beats in a steady rhythm. But in a condition called **Atrial Fibrillation (AFib)**, the top part of your heart starts quivering instead of beating properly. The heartbeat becomes **irregular, fast and uneven**. AFib is not rare, especially in older people, and a lot of people who have it do not even know, because it is often asymptomatic: it has no symptoms.

Why AFib is dangerous: it can cause a stroke

An irregular heartbeat might not sound scary, but here is what can happen:

- When the heart quivers, blood does not get pushed out properly, so some of it **sits still** inside the heart.
- Blood that sits still can turn into a **blood clot**.
- That clot can break loose and travel up to the **brain**.
- It blocks the blood flow, and that part of the brain starts to die. This is called a **stroke**.

A stroke can leave someone paralysed, unable to speak, or in the worst case kill them. Doctors call this kind a **cardioembolic stroke**, because it comes from the heart, and AFib is one of the biggest known causes of it.

The good news is that this is often **preventable**. Doctors can treat AFib once they know about it. The real problem is that people mostly **never find out they have it**.

Why it stays hidden

AFib is sneaky:

- A lot of people feel nothing at all.
- It often comes and goes: it can be there one moment and gone the next.
- It usually happens when you are resting or asleep, which is exactly when nobody is checking.

To catch it, you would have to check the heartbeat **all through the night**. But every device that can do that, a smartwatch, a chest strap or a finger clip, has to be **worn**. People take their smartwatch off to charge it at night, or cannot afford one, or just will not sleep with something on. So the one time it matters most, nothing is watching.

2. What I Learned from Talking to People

To check that this is a real problem and not just my own idea, I talked to people around me about how they keep track of their heart, especially at night.

What I found:

- **Nobody I asked wears anything to bed to check their heart.** A smartwatch owner takes it off to charge overnight, which is the exact time a problem might happen.
- A regular person with a smartwatch most probably **does not check the heart data** and would not know what an irregular rhythm looked like.
- Everyone agreed that chest straps and finger clips are **uncomfortable and easy to forget**.

"It feels uncomfortable on my wrist."

So the problem is clear: heart problems happen at night, but every way of catching them needs you to wear something, and that is the one thing people avoid while they sleep.

3. Looking at the Problem More Closely

Who this affects: everyone, but especially older people, people with heart conditions, and anyone who cannot afford a smartwatch or will not wear one to bed.

Why it keeps happening:

- Dangerous irregular heartbeats like AFib are often silent and come and go, usually at night when no one is watching.
- Every monitor we have needs to touch the body, so it gets taken off, forgotten, or never bought in the first place.
- If it is never caught, an irregular heartbeat can lead to a stroke that could have been prevented.

The missing piece: there is no contactless, effortless, always-on way to watch a heartbeat while someone sleeps.

4. My Innovation Challenge Statement

MY INNOVATION CHALLENGE STATEMENT

How can I keep track of someone's heartbeat all through the night, and warn them if it becomes dangerously irregular, without them having to wear or touch anything at all?

5. What I Built: RadioBeat

Over six weeks I turned the idea into a working prototype. Every time you breathe or your heart beats, your chest moves a tiny amount, and that movement changes the way WiFi signals travel across a room.

RadioBeat reads those changes. The finished prototype can:

- **Measure breathing rate** with nothing worn and nothing touching the person.
- **Estimate heart rate.** This part is experimental, and the app says so (see sections 8 and 9).
- **Measure the gaps between heartbeats** and flag an irregular rhythm like AFib, while refusing to judge when the signal is too noisy.
- **Count how many people are breathing** in range, from their separate breathing rates.
- **Ignore fans and air conditioning** automatically.
- **Run as a web app** with a live view, recorded sessions, settings, and optional sign-in accounts.

6. How It Actually Works

WiFi sensing needs two parts: a transmitter that sends out a WiFi signal, and a receiver that catches it. My receiver reads the Channel State Information (CSI): 64 separate signal channels, about 40 times a second. It sends these to my laptop over USB, where my Python program does the maths.

The hard part is that a heartbeat is tiny. Your chest moves about 5 mm when you breathe, but only about 0.45 mm from a heartbeat, so the heartbeat is almost hidden underneath the breathing. My program works through these steps:

Step	What it does	Why it is needed
1. Clean the signal	Removes the random timing offset and gain jump the radio adds to every packet	Otherwise the radio's own noise is bigger than the chest movement
2. Remove the room	Subtracts the average reflection, keeping only what moved	Walls and furniture reflect far more signal than a chest does
3. Find breathing	Looks for a rhythm between 7 and 36 breaths a minute	Breathing is the biggest movement and must be measured before it can be removed
4. Remove breathing	Works out the breathing wave and its echoes, then subtracts them	The echoes of breathing land right where a heartbeat would be
5. Find the heartbeat	Averages over four minutes, using only channels that agree with each other	The heartbeat is too weak to find reliably in a shorter window
6. Check the rhythm	Measures the time between beats (RMSSD, pNN50, entropy)	AFib shows up as beats with no regular spacing
7. Avoid false alarms	Only warns if five separate checks all pass	Noise can look exactly like an irregular heartbeat

One thing I learned: I cannot just use my laptop's own WiFi. Laptops only give a rough signal-strength number, not the detailed CSI data I need, so I used two small ESP32 chips that give clean signal data and can be placed exactly where I want.

7. Problems I Found and Fixed While Building

Most of my progress came from things going wrong. These were the biggest ones:

What went wrong	Why it happened	What I did
The signal stopped after only 7 packets	The chip only records CSI when WiFi messages are being exchanged, and after connecting it went quiet	The receiver now sends a tiny message to the transmitter 100 times a second to keep the signal flowing
The heart rate showed 70 BPM when my real heart rate was about 100	At 17.5 breaths a minute, the fourth echo of breathing lands on exactly 70 BPM	Worked out breathing and its echoes mathematically and subtracted them
A fan looked like a person breathing	An oscillating fan sweeps at about 0.1 to 0.25 Hz, inside the breathing range	A machine keeps a perfectly steady rate while a person's rate wanders, so perfectly steady signals are removed. In testing the fan was found and breathing recovered at 16.4 against a true 16.8 breaths a minute
It warned of an irregular rhythm in an empty room	Noise made beat gaps that looked like AFib (RMSSD of 309 ms)	Added physical limits no real heart exceeds, and a rule that irregularity must last five readings in a row
After unplugging the board, it kept showing old readings as if they were live	It kept analysing the last 30 seconds of stored data	After 6 seconds of silence it now shows 'sensor offline' and no numbers at all

8. Testing Against a Known Answer

Checking against my own pulse only tells me about one person in one room, and I cannot see what my program is doing wrong. So I built a test program that creates realistic WiFi signals where I already know the right answer, runs them through my real software, and measures how far off it is.

The test copies what the radio really sees: breathing moving the chest by 5 mm, a heartbeat moving it by 0.45 mm, the phase change this causes at 2.4 GHz, reflections from the room, random gain jumps, the chip rounding values to whole numbers, and packets arriving unevenly.

Main results

What I measured	Result
Breathing rate	Within 0.5 breaths a minute
Heart rate using a 30-second window	34 to 51 BPM off: no better than guessing
Smallest chest movement it can pick up	About 1 mm (1.1 BPM error at 1 mm)
Heart rate using a 4-minute window and agreeing channels	Median error 2.7 BPM; 10 out of 16 tests within 5 BPM

Why the heartbeat was hard to find

My first heart-rate results were no better than chance. Even worse, the confidence score gave a room with a heartbeat 3.01 and a room with no heartbeat 3.53, so it could not tell them apart. To find out why, I made the heartbeat bigger and bigger in the test:

Chest movement from the heartbeat	Average error
0.45 mm (a real heartbeat)	34.3 BPM
1 mm	1.1 BPM
2 mm	1.2 BPM
4 mm and above	1.2 BPM

This showed my program was working properly. It picks up movement down to about 1 mm, and a real heartbeat is just below that. So the fix was not a smarter filter: it was collecting more signal.

What improved it

How long it averages	Average error
30 seconds	51.5 BPM
1 minute	28.1 BPM
2 minutes	23.8 BPM
4 minutes	5.5 BPM

Next I changed which of the 64 channels it trusts. It used to pick the loudest ones, but on a weak signal the loudest channel is usually just the noisiest. Now it picks channels whose readings agree with each other. Over 16 tests with different heart rates and room layouts:

Channel choice	Average error	Median error	Within 5 BPM
Loudest channels (before)	21.3 BPM	15.2 BPM	6 of 16
Channels that agree (after)	11.4 BPM	2.7 BPM	10 of 16

Ideas that did not work

- **Removing the room with SVD** made errors worse (13.4 BPM against 8.7), because it removed the moving signal along with the room.
- **Giving heavy weight to heartbeat echoes** made errors worse (12.7 BPM against 10.2), because it kept choosing an echo instead of the real beat.
- **Sending packets faster** did not help, because my program resamples everything to 40 Hz anyway.
- **A lesson about testing:** one early test on only 7 tries said 6.5 BPM. With 16 tries it was 11.4. Small tests can fool you.

Testing the irregular-rhythm check

Test	Result
Normal heart rhythm (3 tests)	All 3 correctly called regular
AFib-like rhythm (3 tests)	All 3 correctly called irregular
Noise designed to look like AFib (4 tests)	All 4 correctly refused to give a verdict
Finding individual heartbeats	37 out of 37 found within 100 ms

IMPORTANT

All of these accuracy numbers come from test signals with a known answer, not from a real person. The next step is testing against a pulse oximeter on real people.

9. Is It Actually Doable? (Being Honest)

In Week 1 I planned to get breathing working first, so that even in the worst case I would still have a working contactless breathing monitor. That plan held up.

- **Breathing is reliable.** It is the solid, working part of RadioBeat.
- **The heartbeat is at the edge of what this hardware can do.** One antenna picks up movement down to about 1 mm and a heartbeat is about 0.45 mm. Averaging longer brings the median error to 2.7 BPM, but about a third of readings are still well off, so the app labels heart rate as experimental.

What RadioBeat cannot do:

- **Replace an ECG.** An ECG reads the heart's electrical signal. RadioBeat reads chest movement.
- **Diagnose anything.** It can only suggest that someone should get properly checked.
- **Cope with a busy room.** Someone walking past moves the signal far more than a heartbeat does.
- **Separate two people breathing at the same rate.** Their signals merge into one.

10. The Parts I Used

Part	What it does	Cost
ESP32-WROOM-32E chip	Transmitter: sends a steady WiFi signal across the bed	\$15
M5StickC Plus2 (ESP32)	Receiver: reads the WiFi signal data (CSI) and sends it to my laptop over USB	\$30
My laptop	Runs my Python signal-processing program and the web app that shows the readings	Varies

Software I wrote: firmware for both chips, the Python signal-processing program, a web app for the monitor (with optional Supabase accounts), and the project website.

11. My Six Weeks: The Plan and What Happened

Week	What I planned	What happened
1	Find the problem and write up my idea	Done: researched the problem and wrote my first proposal
2	Set up both ESP32 chips and get the WiFi signal data onto my laptop	Done: both chips set up and CSI data streaming
3	Detect breathing first and show it as a live wave	Done: live breathing wave on my laptop
4	Pull out the heartbeat by filtering the signal	Done: found and fixed a false 70 BPM reading caused by breathing echoes
5	Spot an irregular rhythm and send a warning	Done: rhythm check with five false-alarm guards, and a web app for the monitor
6	Check my heart rate against a finger sensor, then build the demo	Partly done: built the known-answer test and the demo. Testing against a finger sensor on a real person is still to do

12. What I Would Do Next

- **Test on real people** against a pulse oximeter, over a whole night.
- **Add a second antenna or use a 5 GHz radio.** At 5 GHz the same chest movement causes about twice the signal change, which could lift the heartbeat above the limit I found.
- **Send alerts to family** by text message or phone notification.

13. Why It Would Matter

RadioBeat shows that an important health check could be done with nothing worn, nothing touched, and hardware that costs less than a smartwatch. It could help:

- older people living alone, who could be watched over safely at night;
- families who cannot afford expensive health gadgets;
- anyone with a silent irregular heartbeat that would normally go unnoticed.

In one line: catch the silent heartbeat, prevent the stroke.